

FCUP-CC3042-2024/2025-1S

✓ Concluída

Customizing OpenAI Gym Environments and Implementing Reinforcement Learning Agents with Stable Baselines

Overview

In this assignment, you will develop a reinforcement learning agent using an existing OpenAI Gym / Gymnasium environment as a base. Your task is to introduce specific changes or customizations to the environment and train a reinforcement learning agent using the Stable Baselines library. The goal is to assess how these changes impact the agent's learning process and performance.

Milestones

Week 1: Environment Customization

- **Select an Existing OpenAI Gym / Gymnasium Environment:** Choose an existing environment that interests you. It could be a classic control task, a robotic simulation, or a custom environment.
- **Identify Necessary Modifications:** Carefully analyze the chosen environment and identify areas where you can make meaningful modifications or customizations. These changes could include altering rewards, introducing additional challenges, or modifying the state space.
- **Implement Customizations:** Implement the identified customizations in the environment code. Ensure that the modified environment is compatible with the Gym interface and retains a reasonable level of complexity for reinforcement learning experimentation.

Week 2: Reinforcement Learning Agent

- **Select a Reinforcement Learning Algorithm:** Choose a reinforcement learning algorithm from the Stable Baselines library. Popular choices include Proximal Policy Optimization (PPO), Deep Deterministic Policy Gradients (DDPG), or any other algorithm you find suitable for your customized environment.
- **Implement the Agent:** Write code to create and train a reinforcement learning agent using the chosen algorithm. Make sure to integrate the agent with your customized environment.
- **Hyperparameter Tuning:** Experiment with different hyperparameters to fine-tune your agent's performance. Document the changes you make and the reasons behind them.

Week 3: Evaluation and Analysis

- **Training and Evaluation:** Train your reinforcement learning agent on the customized environment and evaluate its performance using appropriate metrics. Ensure you run multiple experiments to gather statistically significant results.
- **Compare Results:** Compare the performance of your customized environment with the original, unmodified Gym environment. Analyze how your changes affected the agent's learning process and final performance.
- **Prepare a Presentation:** Summarize your findings. Include details about the environment modifications, the reinforcement learning algorithm used, hyperparameter choices, and the comparative results. Discuss the implications of your customizations on the agent's performance.

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◀ Assignment 1: Delivery

Assignment 2 theme definition ▶

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